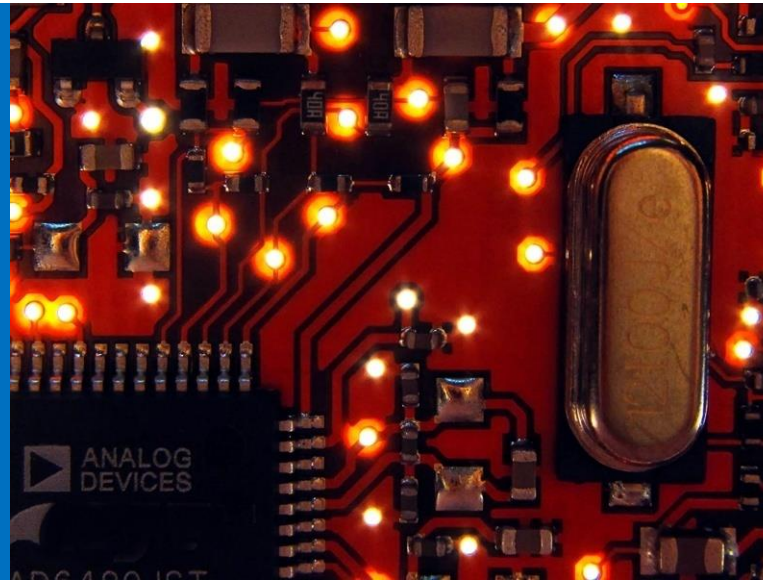
Electronică Analogică

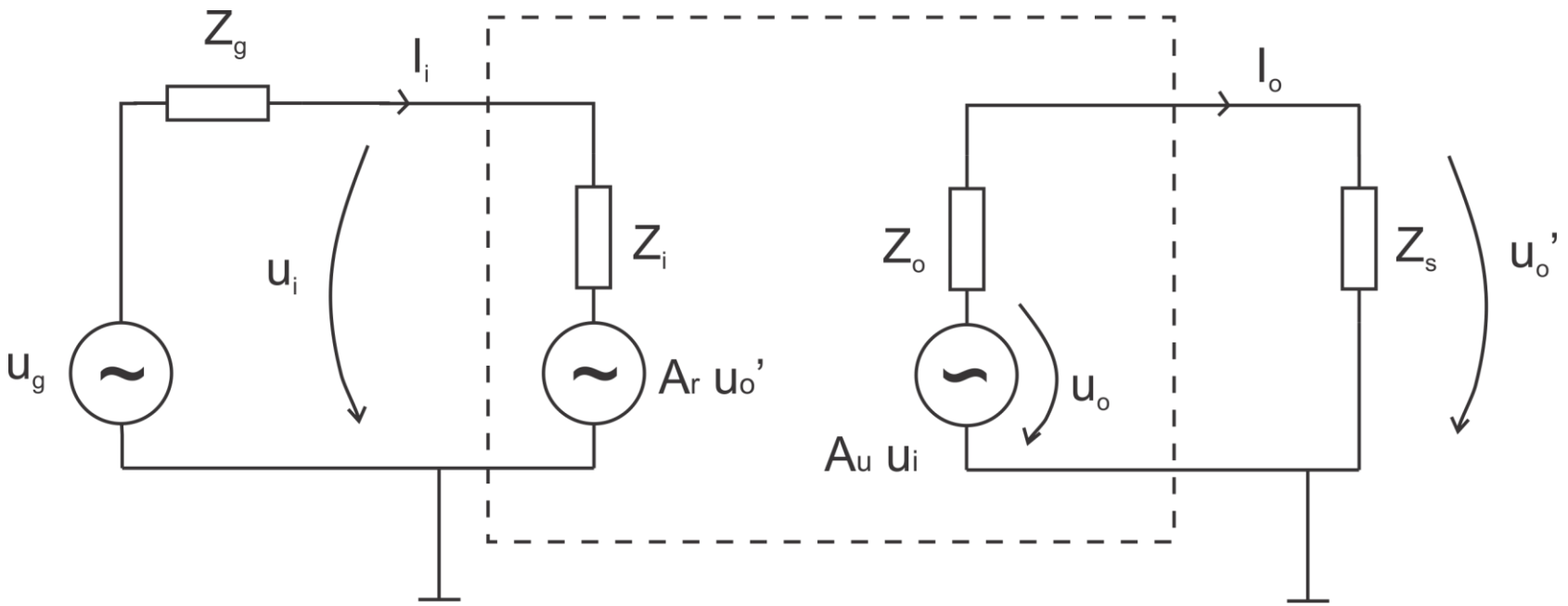


Curs 7 - 8

Amplificatoare (recap)



Model general



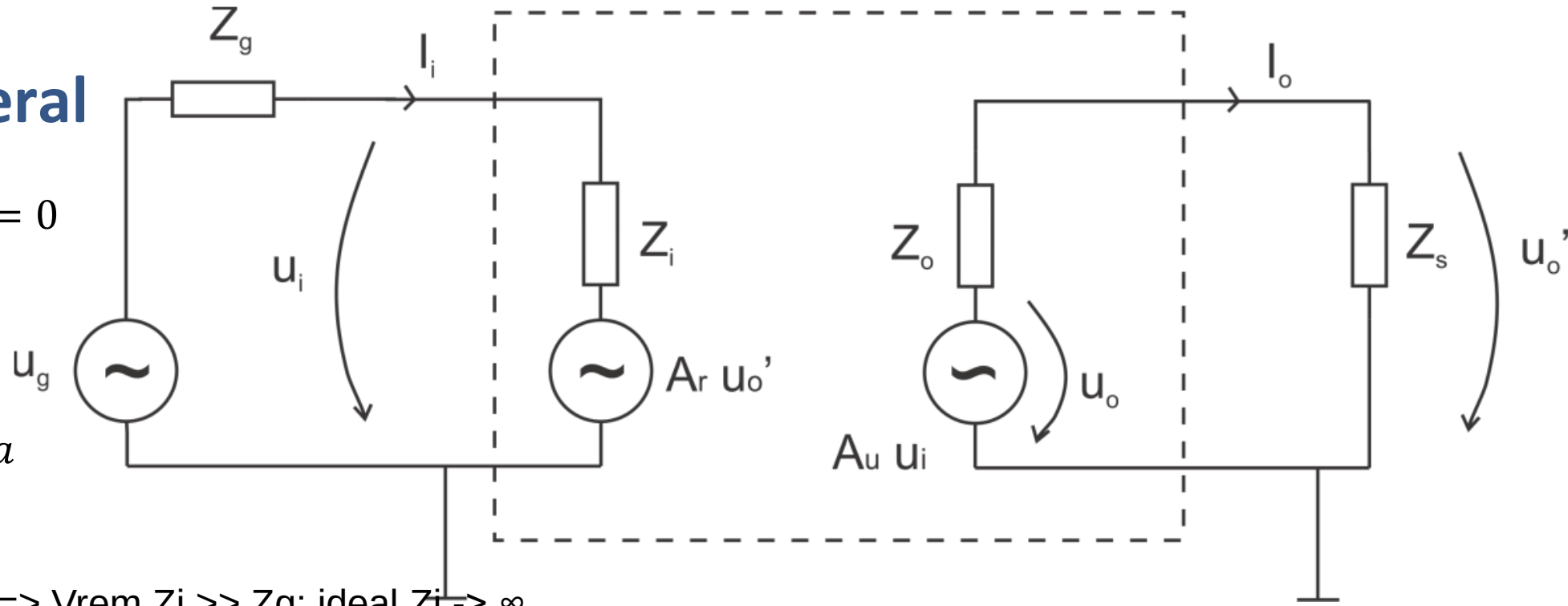
Ar neglijabil

Model general

$$Z_i = \frac{u_i}{i_i} \; pentru \; u_o{}' = 0$$

$$A_u = \frac{u_o}{u_i{}'} \; in \; gol$$

$$A_u{}' = \frac{u_o}{u_i} \; in \; sarcina$$



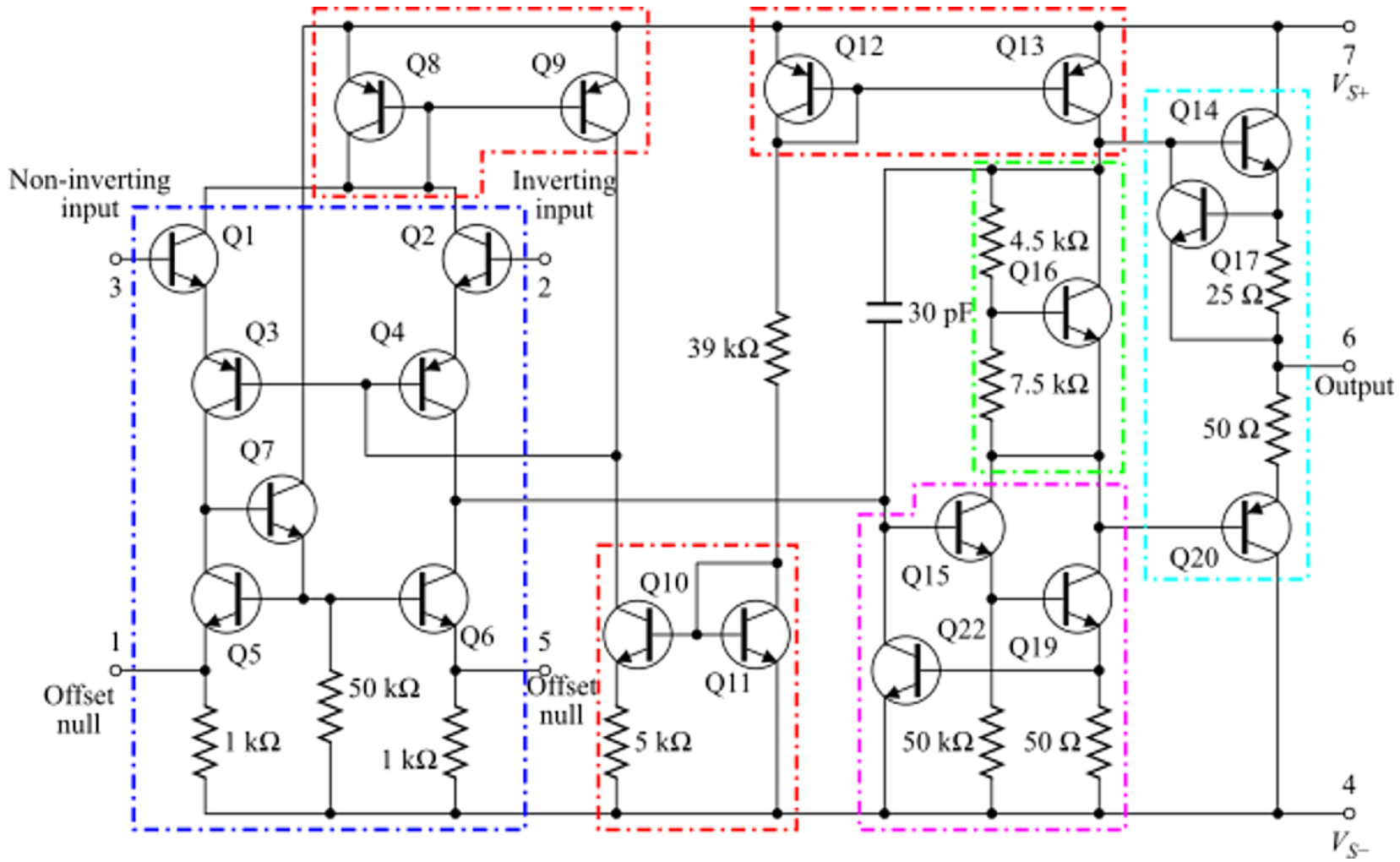
$$u_i = u_g \frac{Z_i}{Z_i + Z_g}$$ => Vrem Zi >> Zg; ideal Zi -> ∞

$$u_o{}' = u_o \frac{Z_s}{Z_s + Z_o}$$ => Vrem Zo << Zs; ideal Zo ->0

$$A_u{}' = A_u \frac{Z_s}{Z_s + Z_o}$$

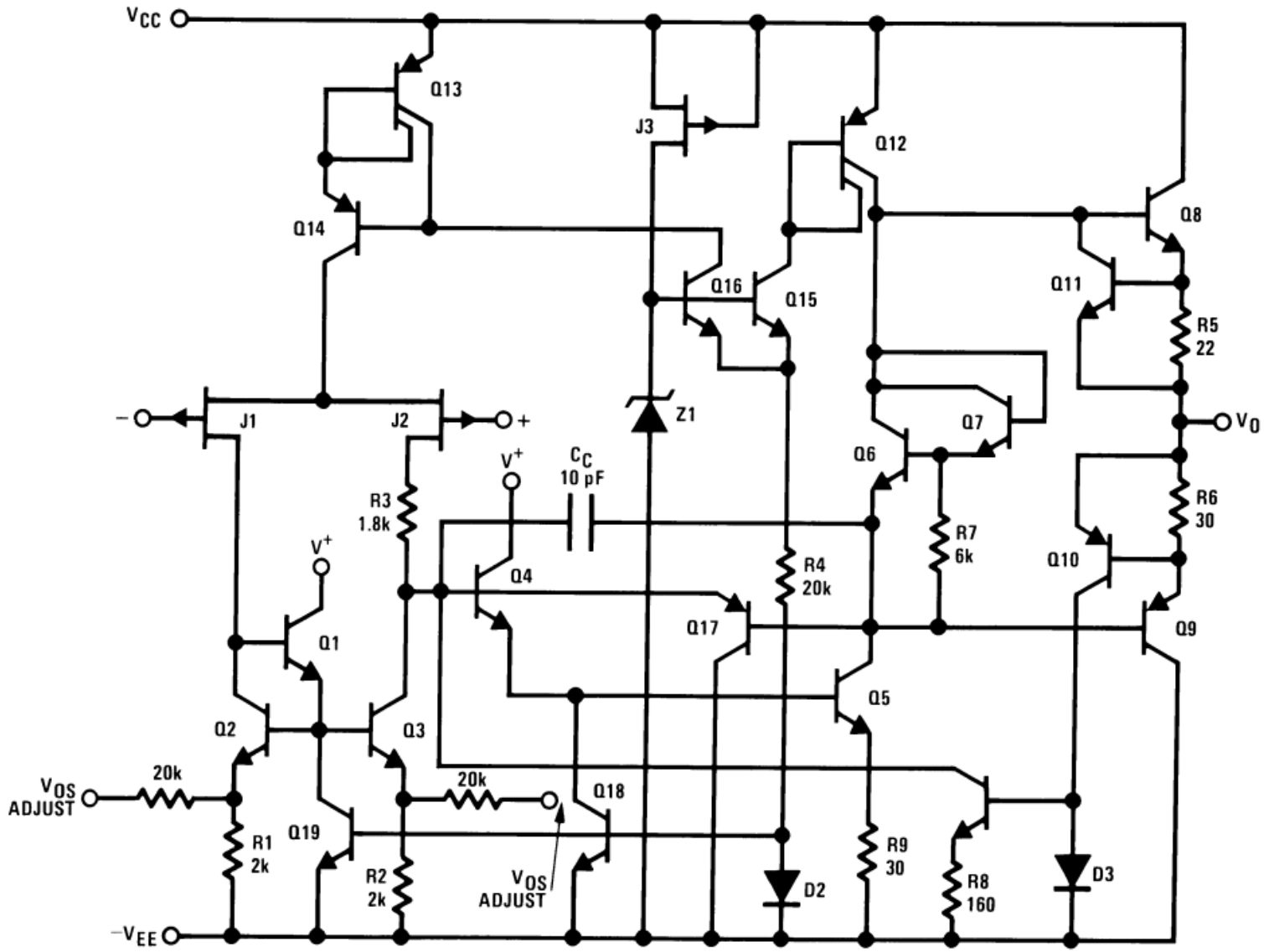
De ce integrate

741



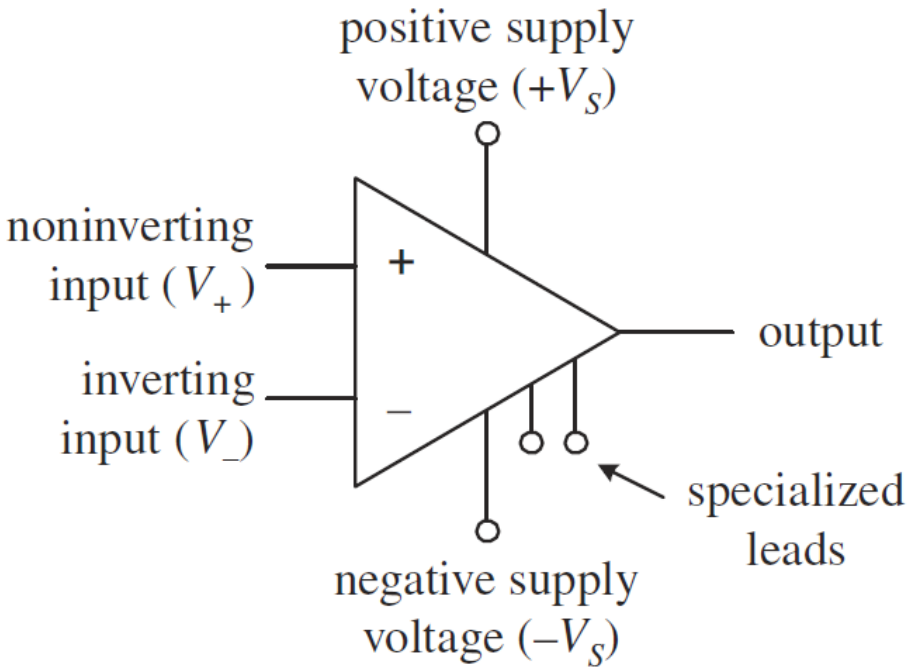
© Daniel Braun

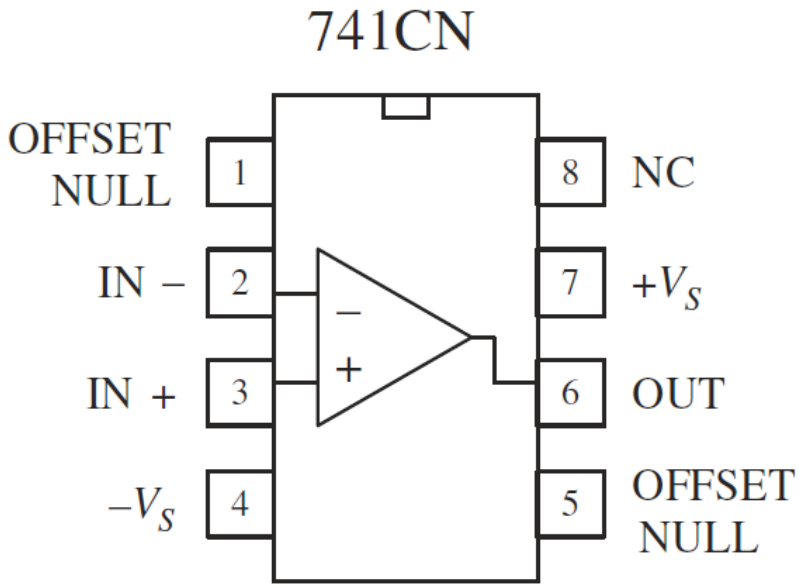
LF411QML-SP



© http://www.ti.com/lit/ds/symlink/lf411qml-sp.pdf

AO – intro



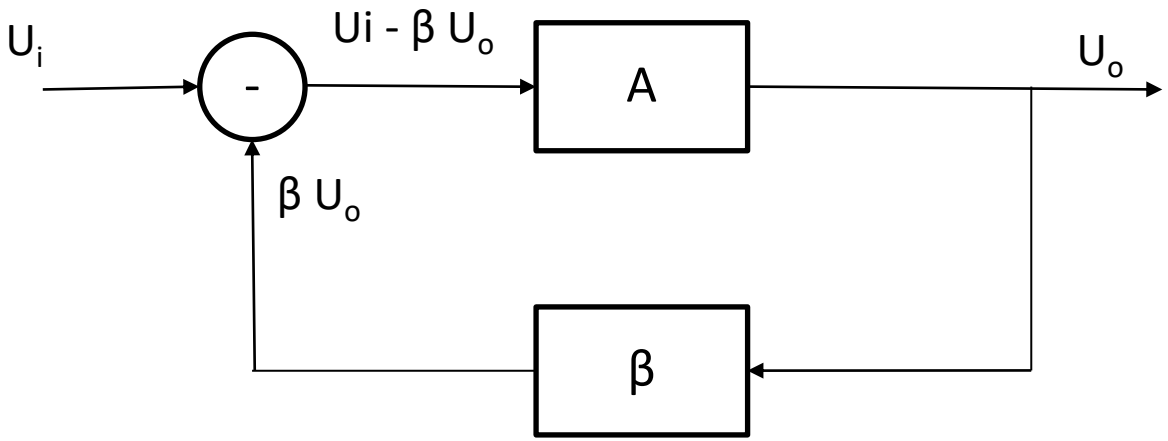


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AO basics



Reactia negativa



$$U_o = A\,(U_i - \beta U_o)$$

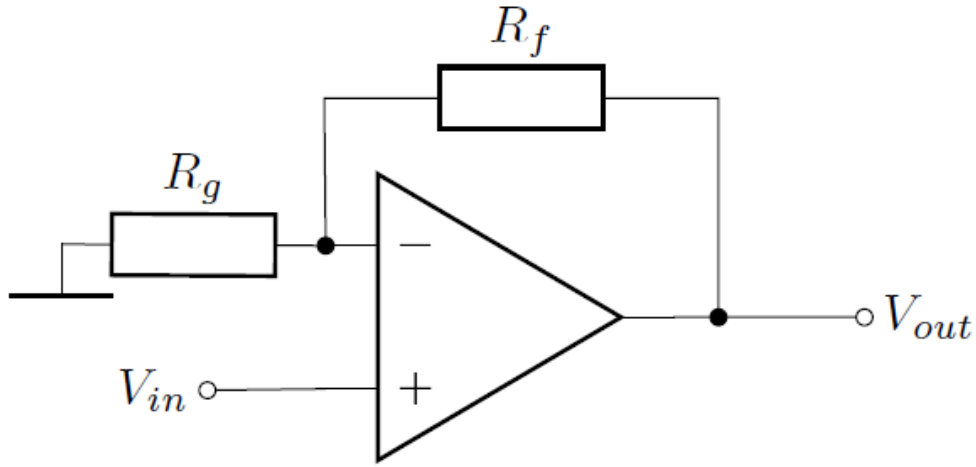
$$U_o(1 + A\,\beta) = AU_i$$

$$U_o = \frac{A}{1 + A\,\beta} U_i$$

$$U_o = \frac{1}{1/A + \beta} U_i$$

Pt A>> => Amplif = 1/ β

AO – cu reactive negativa



Vr = V- = Vout [Rg / (Rg+Rf)]

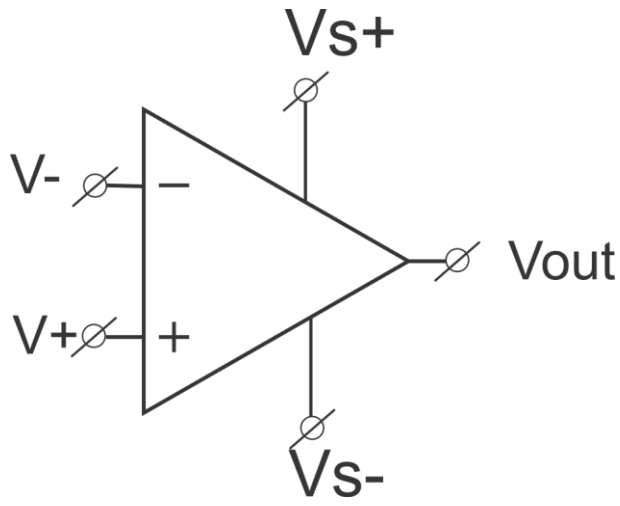
β = [Rg / (Rg+Rf)] => 1/β = [(Rg+Rf) / Rg] = 1 + Rf/Rg
⇒ Amplif = 1+ Rf/Rg

Vo = A x Vi =>
Vin = Vo /A =>
Vin = Vo / (1+ Rf/Rg) =>
Vin = Vo Rg / (Rg+Rf) =>
Vin = V-

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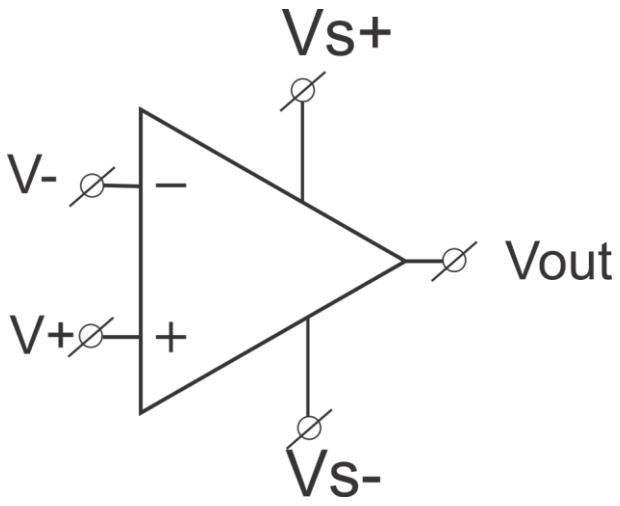
AO – proprietăți

- AO ideal:
  - Vout = A0 (V+ - V-)
  - A0 - infinita
  - dacă avem reacție negativă: v+ = v-
  - i+ = i- = 0
  - Au = ∞ (amplificarea este infinită)
  - Zin = ∞ (impedanță de intrare infinită)
  - Zout = 0 (impedanță de ieșire nulă)

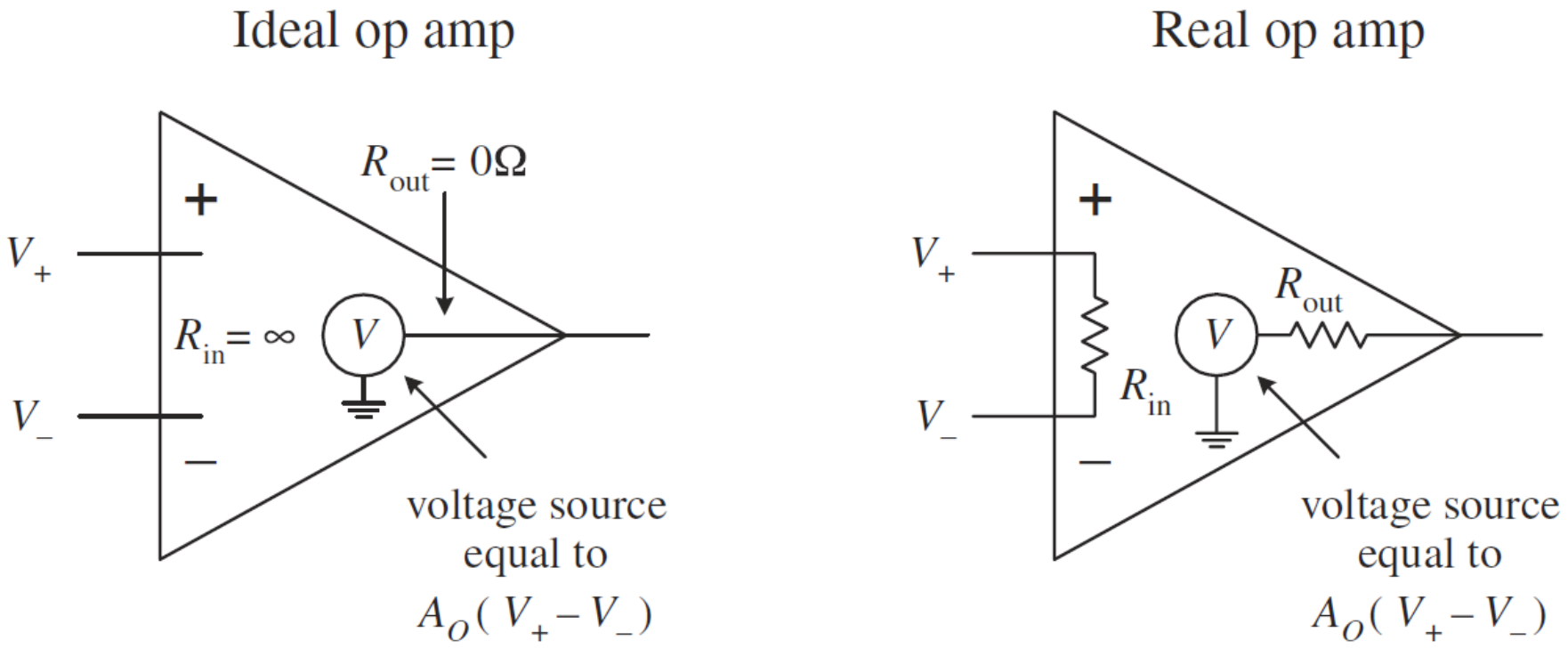


AO – proprietăți

- Obs val reale:
  - Au: 10^4 - 10^6
  - Rin: 10^6 - 10^{12}
  - Iin: 10^{-8} – 10^{-11}
  - v+ != v- (todo v offset)
  - i+ != i- != 0 (to do I offset?)
  - GBP – de introdus si acesta limitare



AO – intro – ideal vs non-ideal



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AO – ex

Circuite liniare cu AO

Amplificatorul inversor

- *Întru-cât avem reacție negativă:*
  - *v+ = v- (1)*
  - *i+ = i- = 0 (2)*
- din (2) => (i+) * R_3=0 => v+ = 0 =>

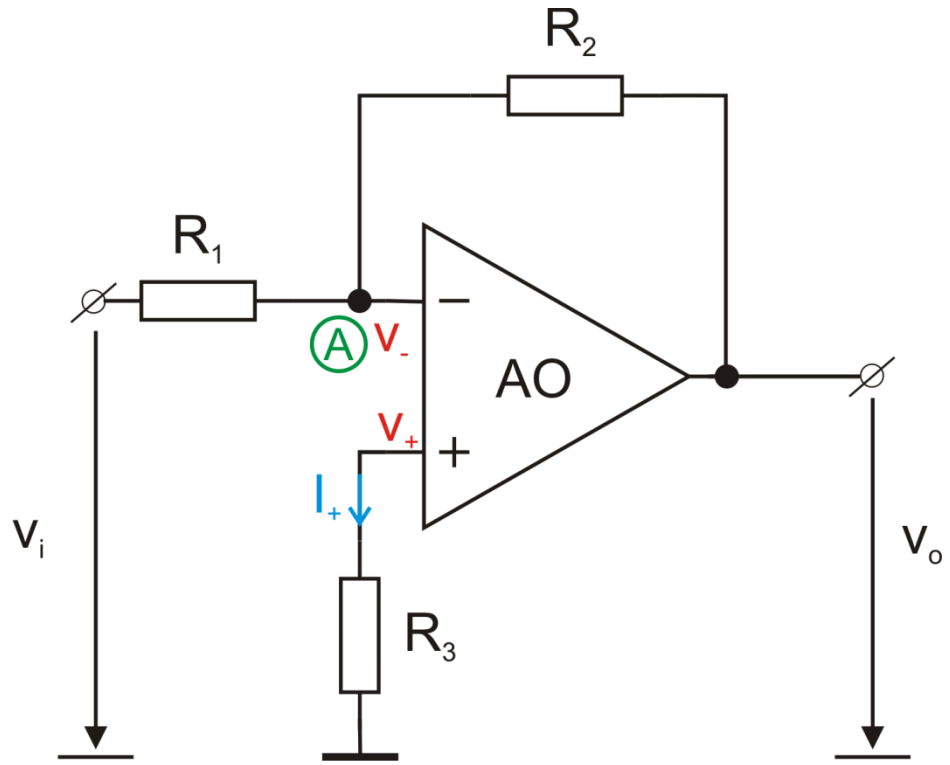
v- = 0 => punctul A este un punct virtual de masă (potențial 0)

- Metoda potențialelor la noduri:

$V(A)\,(1/R_1 + 1/R_2) = v_i/R_1 + v_o/R_2$

$0 = v_i/R_1 + v_o/R_2$

$$\Rightarrow \mathbf{V_o = - v_i * R_2/R_1}$$



Amplificatorul neinversor

- *Se știe că:*
  - *v+ = v- (1)*
  - *i+ = i- = 0 (2)*
- din (2) => (i+) * R_3=0 => v+ = v_i =>

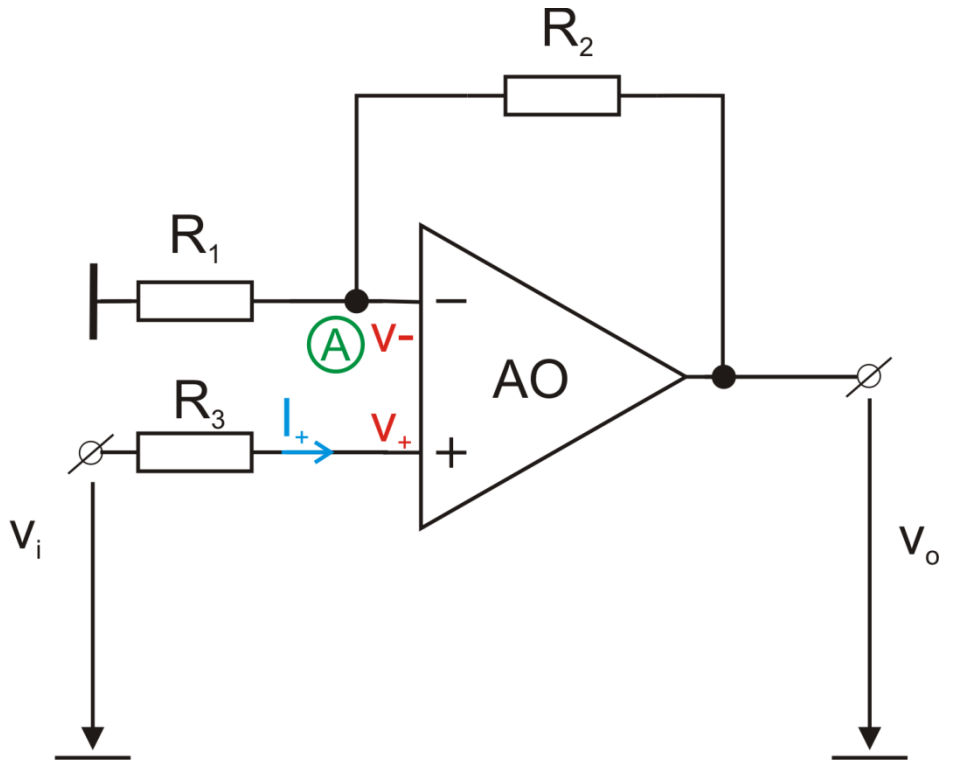
v- = v_i => în punctul A avem v_i

Metoda potențialelor la noduri:

V(A) $(1/R_1 + 1/R_2) = 0/R_1 + v_o/R_2$

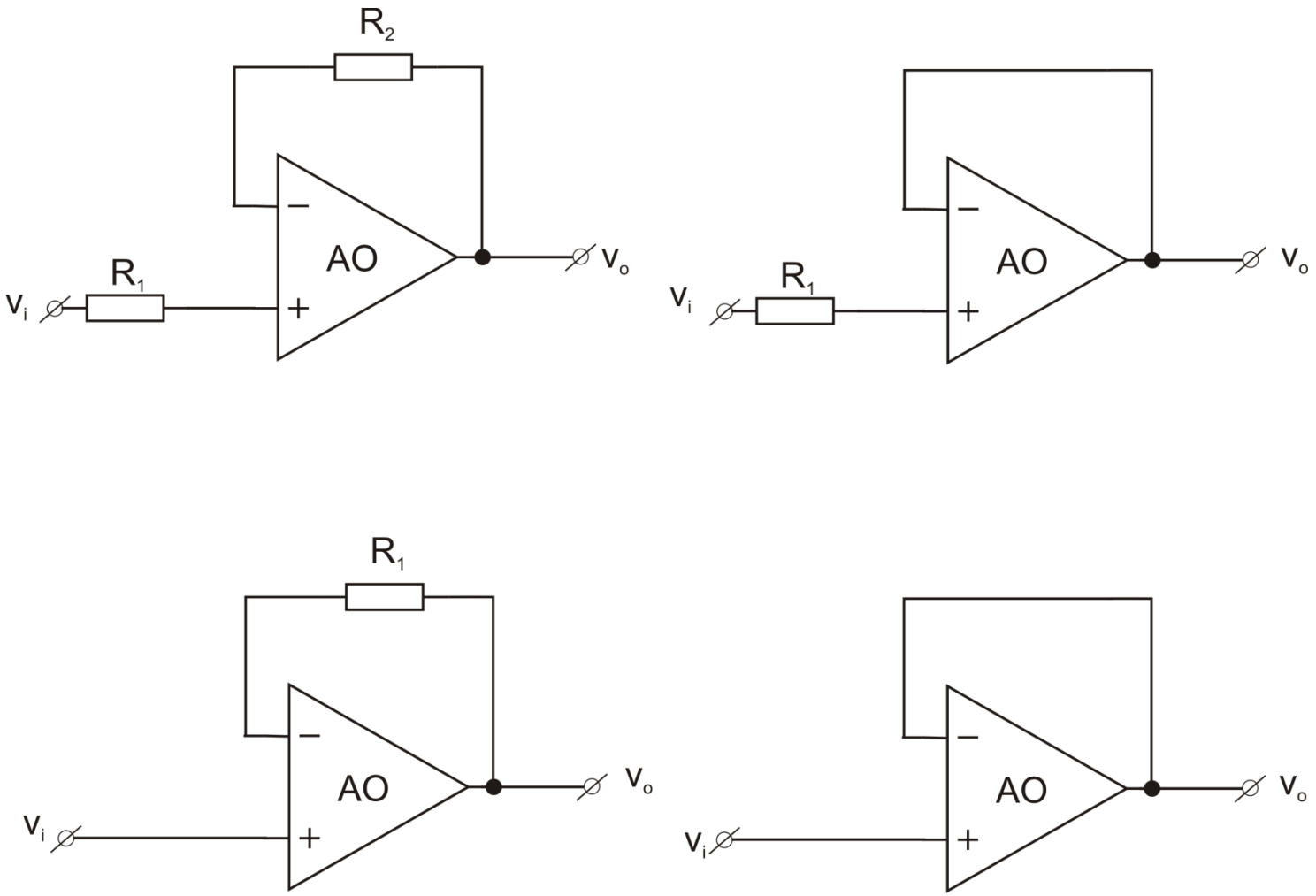
$vi(1/R_1+1/R_2) = v_o/R_2$

=> $\mathbf{v_o = v_i * (1 + R_2/R_1)}$



AO repetor

- $v_o = v_i$ ($A_u = 1$)



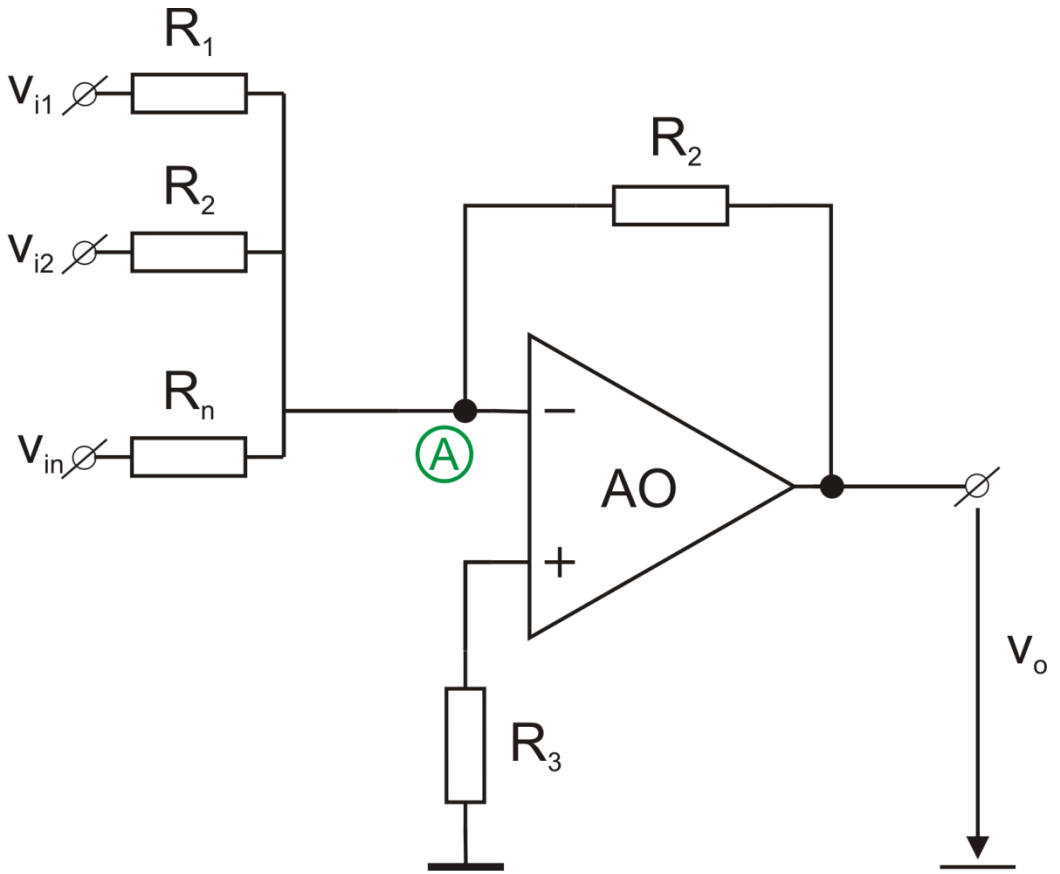
AO sumator inversor

- MPN(A):

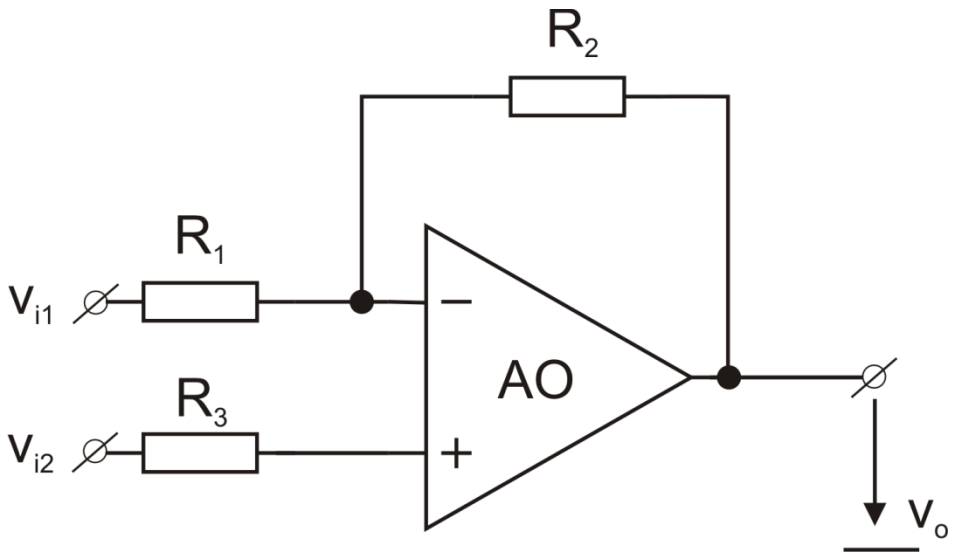
$0 * \text{sum}(1/R_j) = v_0/R_2 +$

$\text{sum}(u_{ij}/R_j),\ j=1,n$

$\Rightarrow v_0 = - R_2 * \text{sum}\ (u_{ij}/R_j),\ j=1,n$



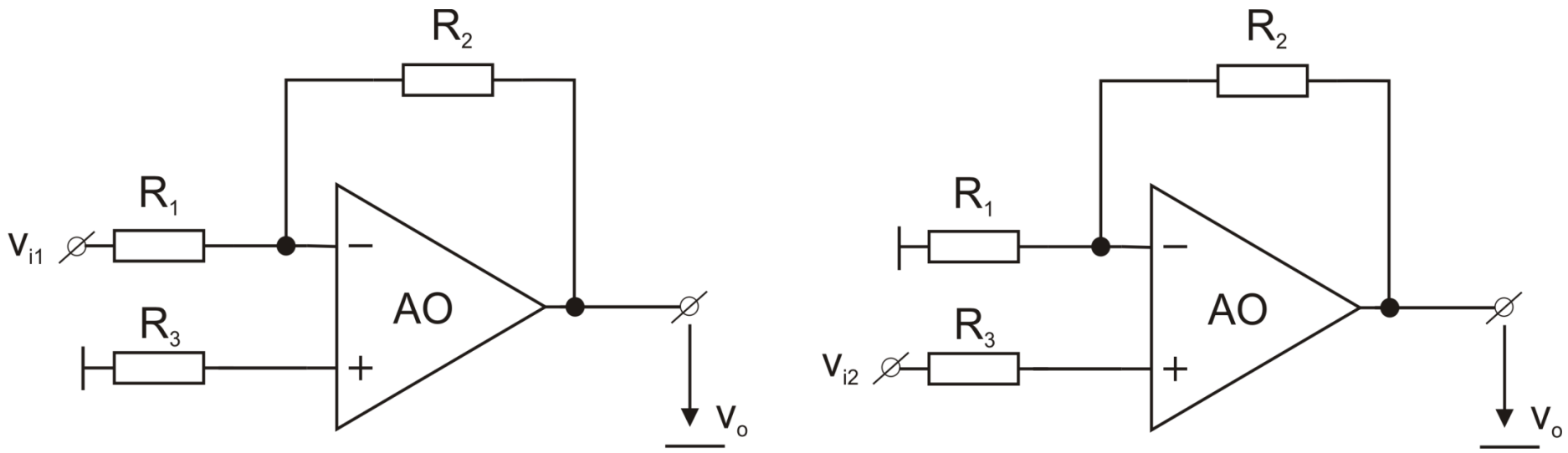
AO – inversor - neinversor



AO – inversor - neinversor

- Metoda superpoziției:

$$v_o = - v_{i1} * R_2/R_1 + v_{i2} * (1 + R_2/R_1)$$



Amplificator diferențial cu AO

- Div tensiune (A): $v+ = v_{i2} * R_4/(R_3 + R_4)$
- MPN (B):

$v+ * (1/R_1 + 1/R_2) = v_{i1}/R_1 + v_o/R_2 =>$

$v_{i2} * R_4/(R_3 + R_4) * (1/R_1 + 1/R_2) =$

$v_{i1}/R_1 + v_o/R_2 =>$

$v_0 = v_{i2} * R_4/(R_3 + R_4) * (1 + R_2/R_1) –$

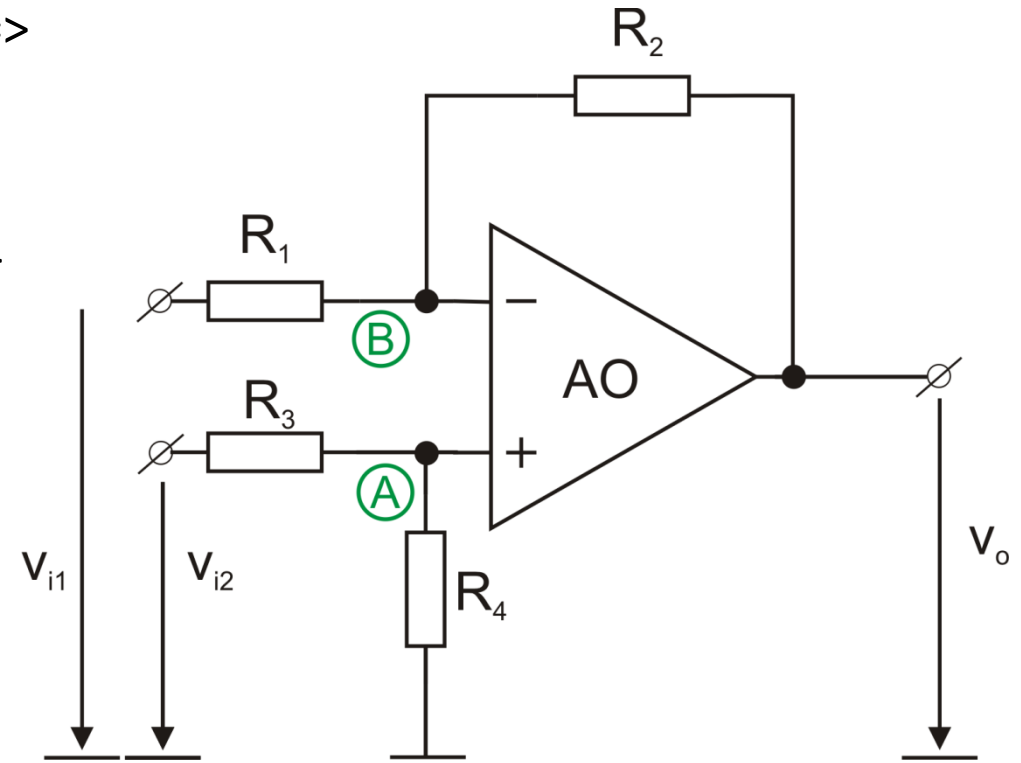
$v_{i1} * R_2/R_1$

- **dacă $R_2/R_1 = R_4/R_3$ =>**

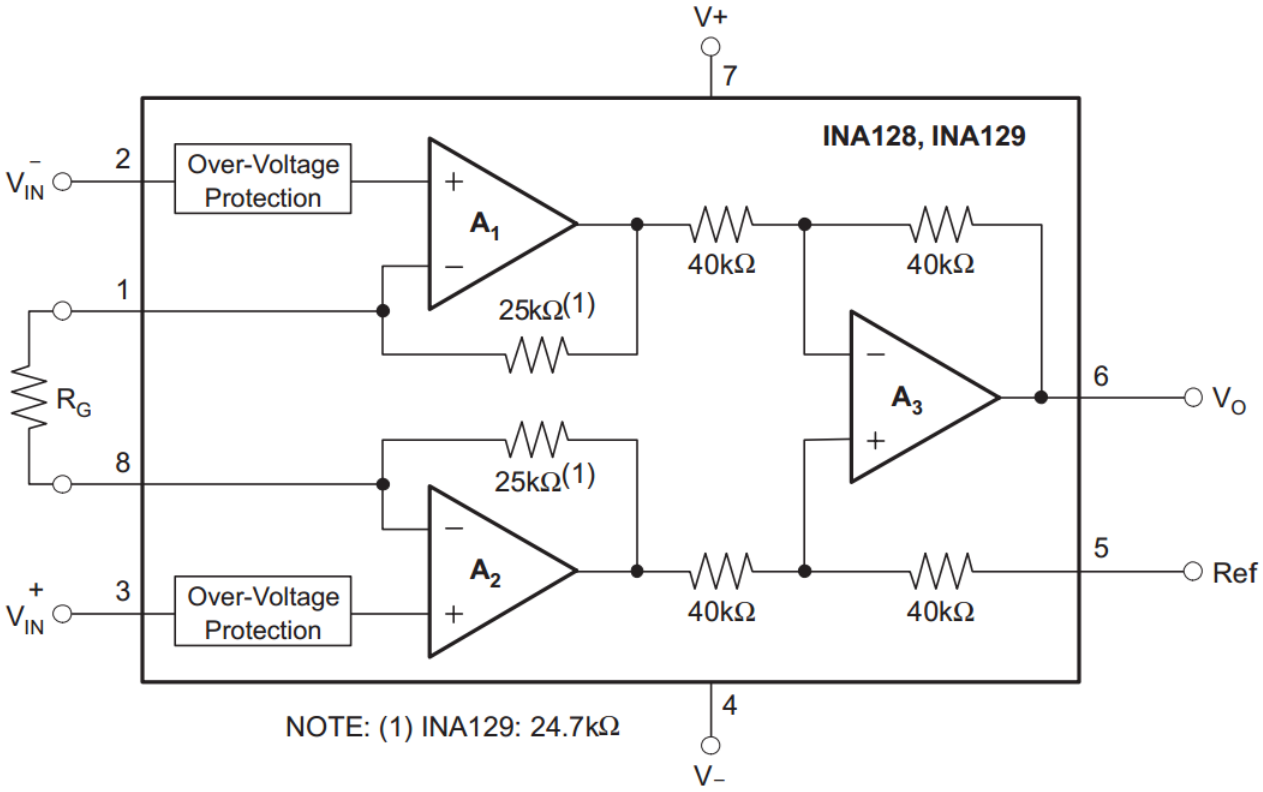
$v_o = R_2/R_1 * (v_{i2} – v_{i1})$

- *dacă $R_4 = R_3$ și $R_2 = R_1$ =>*

$v_o = v_{i2} – v_{i1}$ (scăzător)



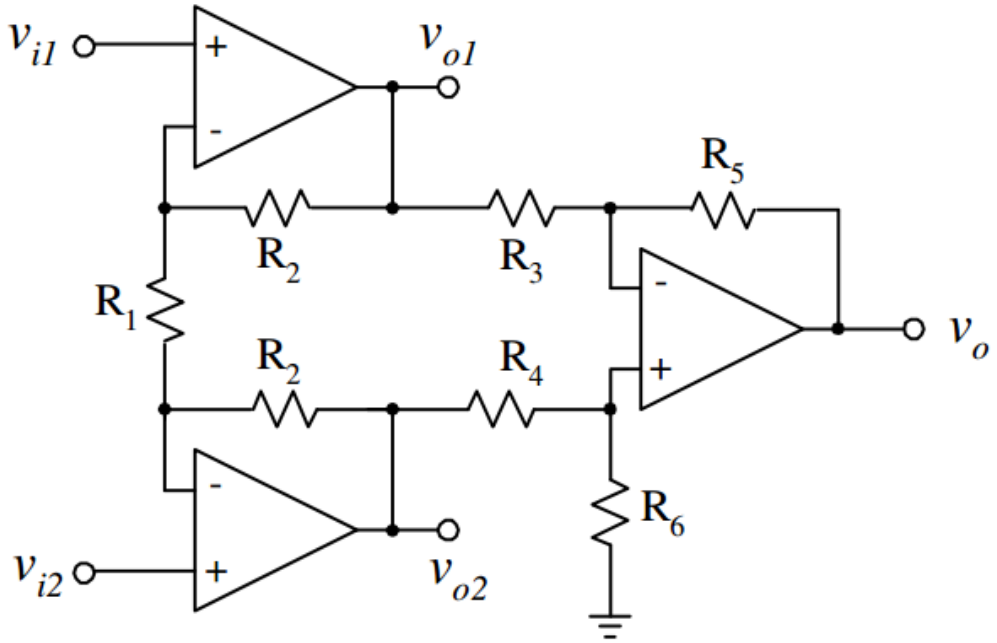
INA



CMMR >>
Rin >>
Low noise

INA

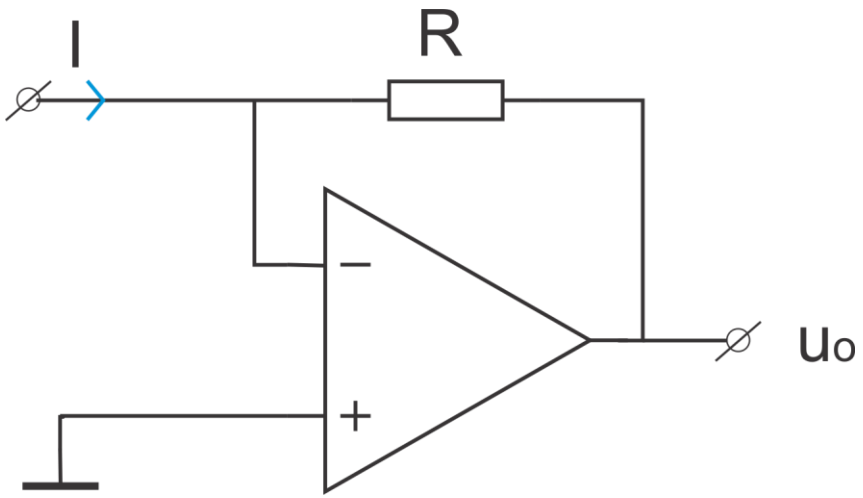
© http://www.ece.tamu.edu/~spalermo/ecen325/Section%20III.pdf



$$v_o = \left(\frac{R_5}{R_3}\right)\left(\frac{R_1 + 2R_2}{R_1}\right)(v_{i2} - v_{i1})$$

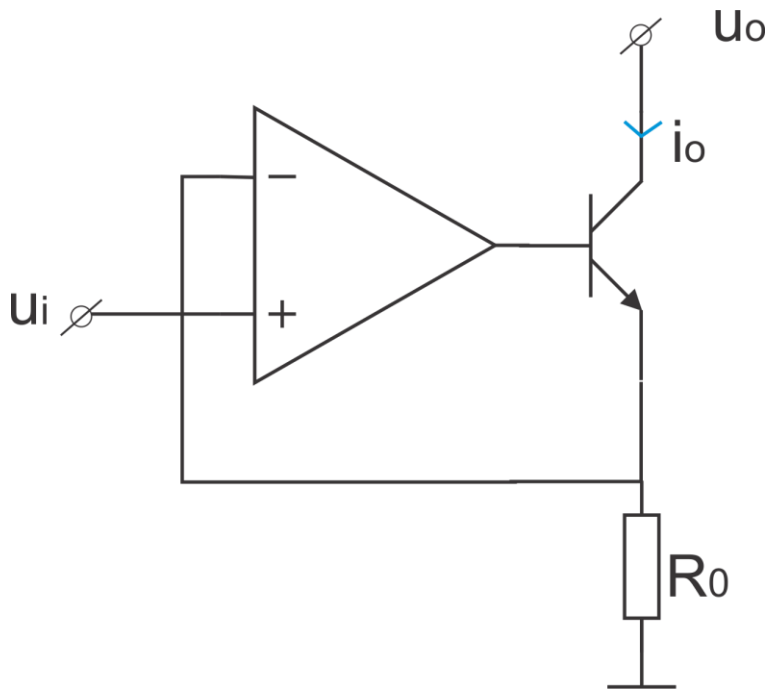
Convertor curent - tensiune

- Uo = -IR



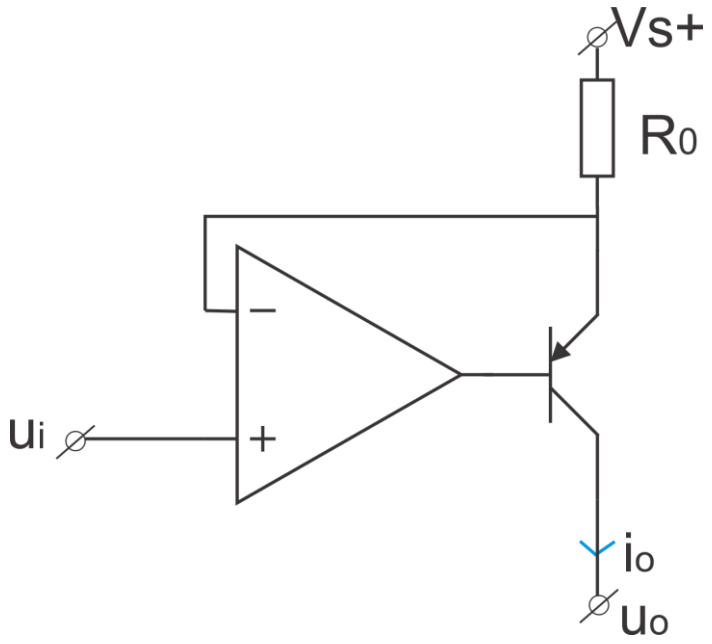
Sursa de curent 1

- Io ≈ ui/R0 (trebuie uo > ui)



Sursa de curent 2

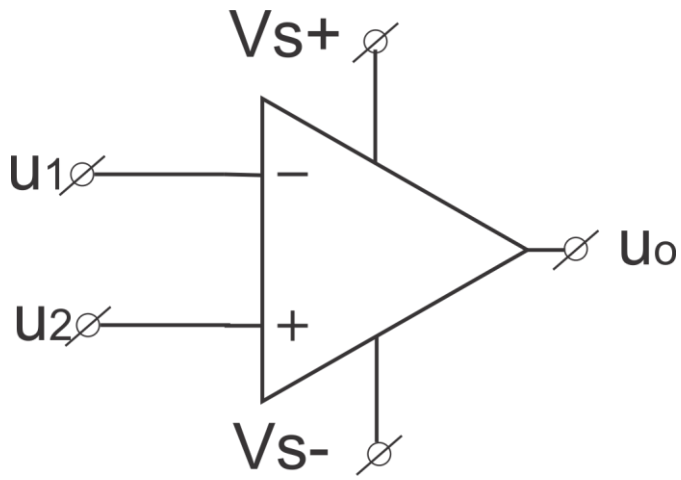
- Io ≈ [Vs+ - (-)ui]/R0 (trebuie uo < ui)



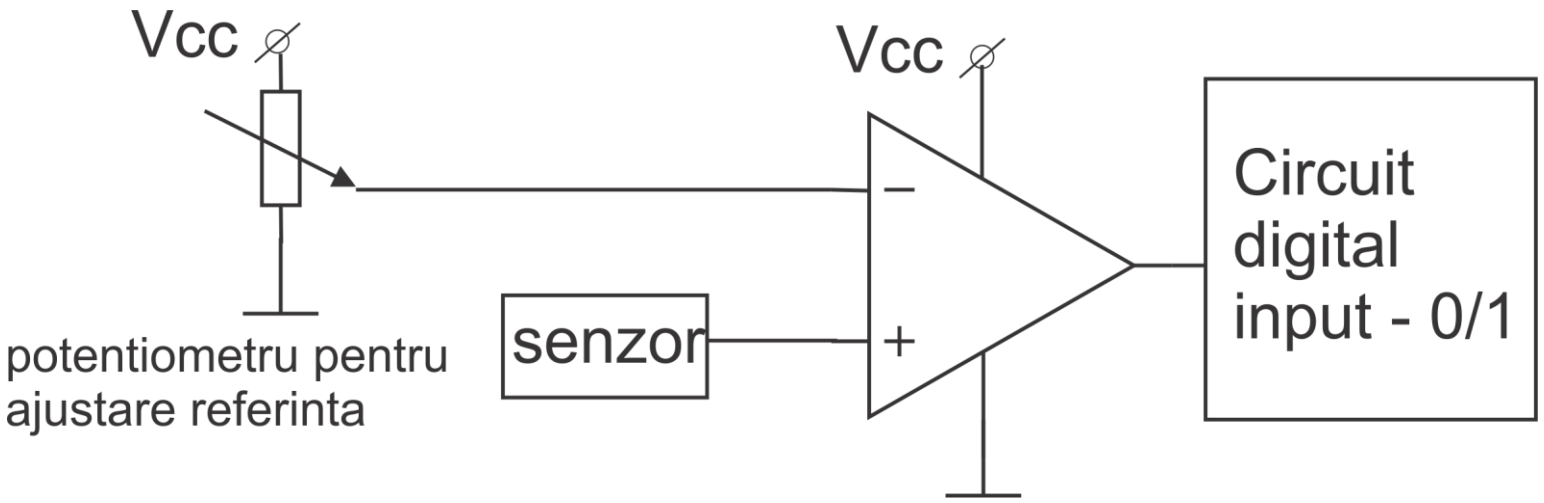
Circuite ne-liniare cu AO

Comparator

- Daca u1>u2 => uo = Vs+
- Daca u1<u2 => uo = Vs-

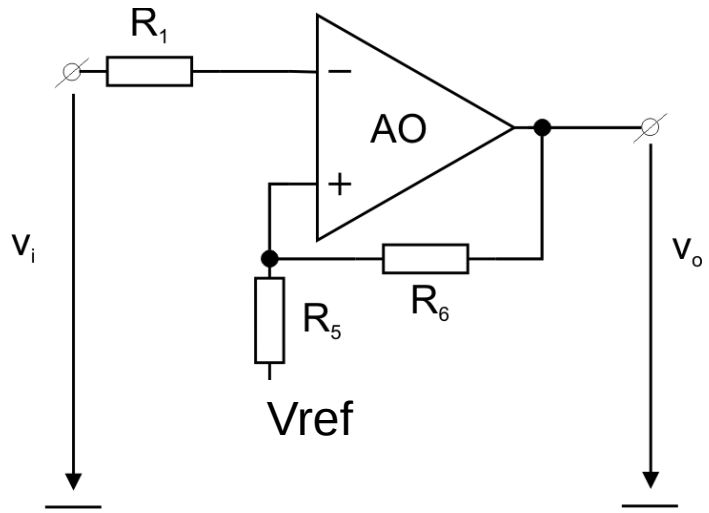


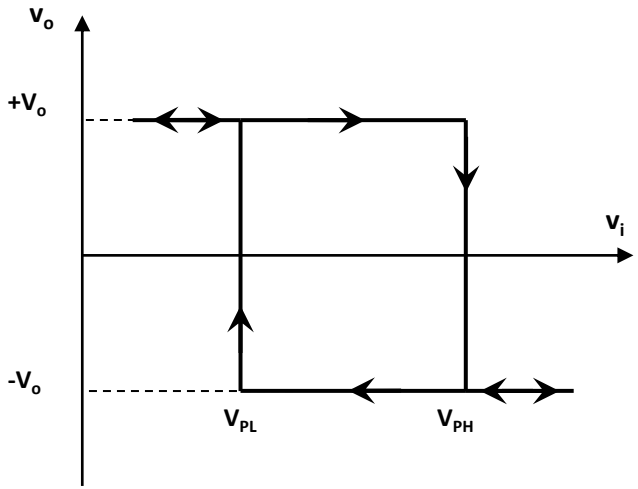
Comparator - aplicatie



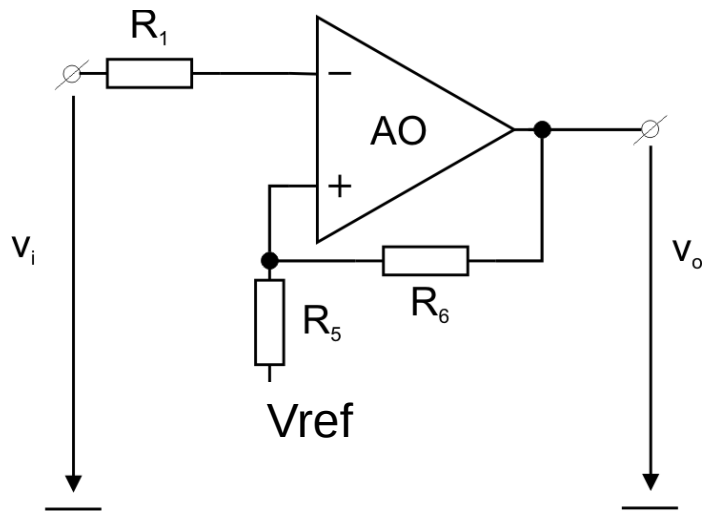
Exemplu, expl, simulare (la curs)

Cumparator cu histerezis





Cumparator cu histerezis



V+ din MPN

2 cazuri:

V+ > Vi

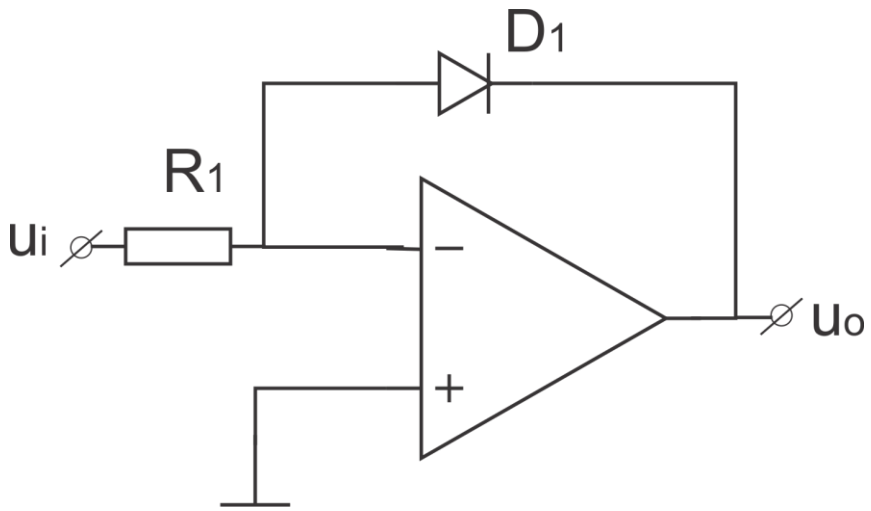
V+ < Vi

simulare (la curs)

Logaritmare

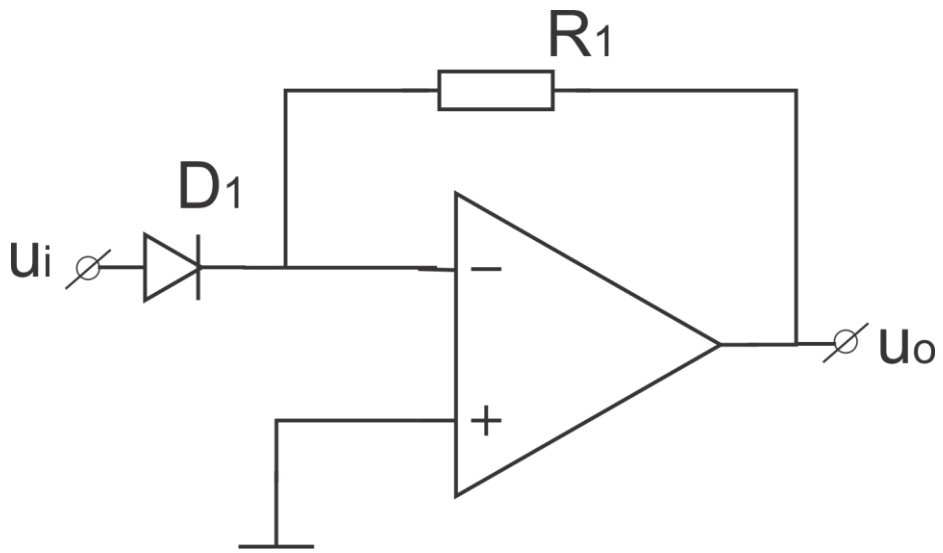
- uo = -q/kt ln (v_{in}/ I_0R_1)

$$i_D = I_0 e^{\frac{qu_D}{kT}}$$

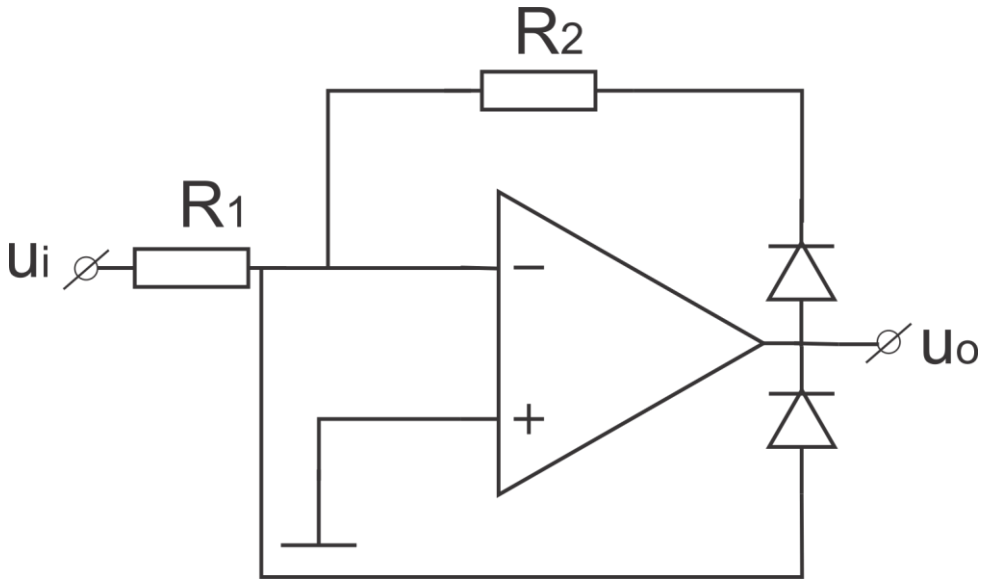


Exponentiere

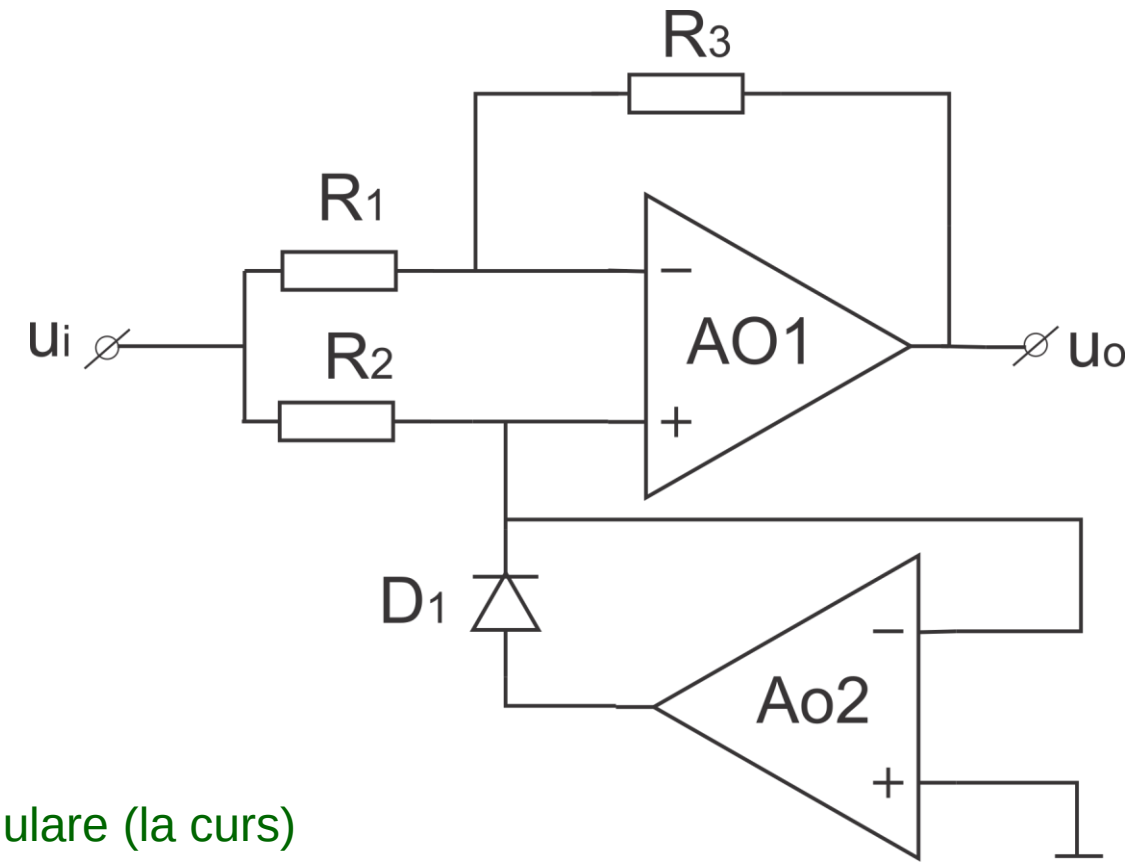
- Vo = -R Io e^(q*ui/kt)



Redresor – mono-alternanta



Functia modul (redresor ideal – bi-alternanta)



Dem + simulare (la curs)

Functia modul (redresor ideal – bi-alternanta)

- //demonstrație la curs

Ex OPAMPS

- LM741 / RC4558
- LM324
- LM358
- LMV358
- TL071
- NE5532
- OP007
- OPA277
- OPA192